

there has a little error at the thrid paragraph's begin:

"Figure 23.7.2" → "Figure 23.8.2"

Paper Review

Title: An 88.36TOPS/W Bit-Level-Weight-Compressed LLM Accelerator with Cluster-Aligned INT-FP-GEMM and Bi-Dimensional Workflow Reformulation

Conference: ISSCC 2025 February 16-20

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Summary

This paper addresses three major challenges in efficient deployment of large language models (LLMs) by proposing a novel accelerator architecture. The key innovations include:

1. **ABAF (Active-Bit-Allocate Format):** The paper introduces a compression format for weights and KV-cache, along with a Prime-Number-Based Calculator (PNBC) for fast decoding without requiring a lookup table. This design effectively reduces the data transfer and storage overhead caused by frequent EMA access during LLM inference.
2. **CAMP-PE (Cluster-Aligned Mixed-Precision Processing Element):** To address the low energy efficiency of mixed-precision GEMM operations, the authors propose a clustering approach that converts floating-point operations into integer computations by grouping activations by exponent, significantly improving efficiency.
3. **BMSR (Bi-Dimensional MAC-based SFU Reformulation):** To reduce SFU bottlenecks in nonlinear operations, the paper reformulates nonlinear functions as MAC operations, allowing them to be processed directly within PEs and increasing overall hardware utilization.

Strengths

- **ABAF** presents an elegant integration of compression, decoding, and dataflow reformation, effectively addressing the high EMA cost in autoregressive models.
- **CAMP-PE** simplifies GEMM operations by transforming activation computations into integer operations using clustering.
- **BMSR** smartly converts nonlinear functions into linear MACs, allowing these operations to be executed in PEs and bypassing SFU limitations.
- **Optimization targeting LLM model deployment**, addressing a key issue in current AI development.
- **Strong persuasiveness** from design concept to actual testing.

- **Circuit architecture is problem-oriented with clear objectives.**

Weaknesses

- The paper lacks sufficient evidence to support the claim that **AWSC** (Active Word Shift Compression) does not affect model accuracy. Bit-shifting operations in deep learning can introduce rounding or truncation errors. The paper does not specify how many bits are used for intermediate values or whether saturation/clipping mechanisms are applied.
- The use of **Taylor expansion** and **Horner's method** for nonlinear function approximation lacks detailed error analysis.
- The paper does not discuss adaptability to bit-widths other than **8 bits**, raising concerns about generality.
- Evaluation on larger models such as **GPT-3** or **LLaMA-13B** is absent. It is unclear whether the architecture can maintain accuracy or avoid performance bottlenecks like **I/O bandwidth** or increasing **EMA** with longer context lengths.

Suggestions for Improvement

- Include **precision error analysis** and evaluate the accelerator's scalability under higher sparsity.
- Provide **performance results on larger-scale LLMs** to assess generalization and robustness.

Overall Evaluation

This paper presents a strong **hardware-software co-design** addressing real-world bottlenecks in LLM inference. With additional **precision analysis** and **generality validation**, it has the potential to become a valuable reference for both edge and cloud LLM accelerator designs.

论文评审

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简要总结

本论文针对当前LLM高效部署所面临的三个主要挑战提出了相应的解决方案，提出了一个针对LLM模型的高

效加速电路，主要创新性成果有：

1. **ABAF**: 本文提出了一种对权重和KV缓存进行压缩的方式 (Active-Bit-Allocate-Format)，同时针对这种压缩方式的快速解码提出了 Prime-Number-Based-Calculator，这是一种基于质数的计数器，可以实现不需要查找表的高效解码。这两者的提出有效解决了LLM推理过程中频繁进行EMA访问导致的数据传输和存储成本过高的问题。
2. **CAMP-PE**: 本文提出了一种新的计算方法: Cluster-Aligned Mixed-Precision Processing Element 用以解决传统LLM推理过程中混合精度矩阵计算能效较低的问题。通过将activation以指数分群，将浮点数计算转为整数计算实现更高效率的计算。
3. **BMSR**: 为了解决传统LLM推理过程中非线性函数计算过于依赖SFU导致电路空闲，利用率低等问题，通过BMSR将非线性计算转化为MAC计算在PE中完成，提高电路工作效率。

主要成果与优点

- ABAF设计巧妙，结合了数据压缩，解码和数据流重构，有效解决了自回归模型中频繁EMA的问题。
- CAMP-PE用分群并将activation计算转化INT计算的方法简化了GEMM计算。
- BMSR巧妙的将非线性计算转化为线性MAC计算，并且能够直接在PE中完成计算，绕过了SFU瓶颈。
- 面向LLM模型部署的优化，解决AI发展热点问题。
- 从设计思路到实际测试，说服力强。
- 电路架构设计针对具体问题，目标明确。

不足之处

- 论文在论证关于AWSC不影响精度的部分时证据不足。在深度学习模型中对齐或移位操作是十分敏感的，对weight进行移位虽然ABs没有改变，但是没有论证是否存在舍入误差或是截断风险，未明确说明使用多少bit表示计算结果，是否设计了饱和截断等保护机制。
- 使用Taylor展开和horner法则简化非线性计算时缺乏精度误差分析。
- 对8bits 以外的位宽格式适应性没有展开讨论，影响电路的通用性论证。
- 未对较大模型如GPT-3， LLaMA-13B进行测试，不确定该电路是否会在较大模型上出现精度损失，热点转移的问题（比如计算瓶颈从SFU变为IO带宽），上下文变长导致EMA再次升高。

改进建议

增加对精度误差的分析，分析在稀疏性更高的情况下电路的扩展适应能力；补充对较大模型的测试结果。

总评

本文是一项针对实际LLM推理瓶颈的优秀硬件-算法协同设计工作。若能进一步补充精度分析和通用性评

估，有潜力成为边缘或云端LLM加速器设计的重要参考方案。